Project Report

Spring 2025

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Project: Contacts List #2

Note: As you write each section, try to be as clear and detailed as possible. Your goal is to communicate your thought process and work clearly. Don't worry if you faced challenges or made mistakes; discussing these is a valuable part of learning and shows your problem-solving skills! Remember, there's no single 'right' way to do these tasks, so be creative and honest in your responses.

Problem Statement (2-3 Paragraphs):

The goal was to create a contact management system where users can add, edit, delete, search, import, and export contacts through a menu system interface. The program outputs contact information in a formatted table with name, birthdate, age, and phone number. The main challenges were validating dates in multiple formats, formatting phone numbers correctly, and organizing contacts properly using the binary search tree.

A key requirement was implementing a BST data structure to store contacts sorted alphabetically by last name and first name, instead of using a simple list. This makes sure the program scales well and teaches important computer science concepts about efficient data organization.

Design (1-3 Paragraphs):

We structured the program with four classes (contact, BST, Data manager, and main). The BST was the main design choice because it automatically kept contacts sorted and was required. When adding contacts, I compare their keys to know where to be placed. The traverse_in_order() method returns them in order.

Testing (1-2 Paragraphs + screenshots of 3 test cases):

I tested by adding contacts and verifying they displayed properly in alphabetical order with the proper format. The phone number formatting forked for 10 digit numbers and shorter numbers were shown normally. Date validation rejects improper formats and accepts the yyyy-mm-dd format. The search function properly filters contacts. All the menu options work without problems and exited as expected.

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HW4 – Contact List, Version 2
Solution by CodePuppies

1. Enter a new Contact from the console
2. Import Contacts from a file
3. Display all Contacts on the console
4. Export all Contacts to a file
5. Delete a single Contact
6. Edit a single Contact
7. Display a selected subset of Contacts on the console
8. Export a selected subset of Contacts to a file
9. Exit
Enter choice: 1

Enter first name: Sarah
Enter last name: Johnson
Enter birthdate (m/d/y or yyyy-mm-dd): 1992-12-20
Enter phone (digits and + allowed): 1234567890
Contact added.

1. Enter a new Contact from the console
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6. Edit a single Contact
7. Display a selected subset of Contacts on the console
8. Export a selected subset of Contacts to a file
9. Exit
Enter choice: 5

Enter first name to delete: Sarah
Enter last name to delete: Johnson
Contact deleted.

1. Enter a new Contact from the console
2. Import Contacts from a file
3. Display all Contacts on the console
4. Export all Contacts to a file
5. Delete a single Contact
6. Edit a single Contact
7. Display a selected subset of Contacts on the console
8. Export a selected subset of Contacts to a file
9. Exit
Enter choice: 3

No contacts.

1. Enter a new Contact from the console
2. Import Contacts from a file
3. Display all Contacts on the console
4. Export all Contacts to a file
5. Delete a single Contact
6. Edit a single Contact
7. Display a selected subset of Contacts on the console
8. Export a selected subset of Contacts to a file
9. Exit
Enter choice: 9

HW4 Complete
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Solution by CodePuppies

1. Enter a new Contact from the console
2. Import Contacts from a file
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5. Delete a single Contact
6. Edit a single Contact
7. Display a selected subset of Contacts on the console
8. Export a selected subset of Contacts to a file
9. Exit
Enter choice: 1

Enter first name: Alice
Enter last name: Smith
Enter birthdate (m/d/y or yyyy-mm-dd): 1995-03-10
Enter phone (digits and + allowed): 5559876543
Contact added.

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5. Delete a single Contact
6. Edit a single Contact
7. Display a selected subset of Contacts on the console
8. Export a selected subset of Contacts to a file
9. Exit
Enter choice: 1

Enter first name: Bob
Enter last name: Jones
Enter birthdate (m/d/y or yyyy-mm-dd): 1988-07-22
Enter phone (digits and + allowed): 4155551234
Contact added.

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5. Delete a single Contact
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8. Export a selected subset of Contacts to a file
9. Exit
Enter choice: 1

Enter first name: Charlie
Enter last name: Brown
Enter birthdate (m/d/y or yyyy-mm-dd): 1992-11-05
Enter phone (digits and + allowed): 2125553456
Contact added.

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5. Delete a single Contact
6. Edit a single Contact
7. Display a selected subset of Contacts on the console
8. Export a selected subset of Contacts to a file
9. Exit
Enter choice: 3

All Contacts

Last Name, First Name        Birthdate    Age Phone
Brown, Charlie               1992-11-05    33 +1 (212) 555-3456
Jones, Bob                   1988-07-22    37 +1 (415) 555-1234
Smith, Alice                 1995-03-10    31 +1 (555) 987-6543


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4. Export all Contacts to a file
5. Delete a single Contact
6. Edit a single Contact
7. Display a selected subset of Contacts on the console
8. Export a selected subset of Contacts to a file
9. Exit
Enter choice: 9

HW4 Complete
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HW4 – Contact List, Version 2
Solution by CodePuppies

1. Enter a new Contact from the console
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8. Export a selected subset of Contacts to a file
9. Exit
Enter choice: 1

Enter first name: John
Enter last name: Doe
Enter birthdate (m/d/y or yyyy-mm-dd): 1990-05-15
Enter phone (digits and + allowed): 5551234567
Contact added.

1. Enter a new Contact from the console
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3. Display all Contacts on the console
4. Export all Contacts to a file
5. Delete a single Contact
6. Edit a single Contact
7. Display a selected subset of Contacts on the console
8. Export a selected subset of Contacts to a file
9. Exit
Enter choice: 3

All Contacts

Last Name, First Name        Birthdate    Age Phone
Doe, John                    1990-05-15    35 +1 (555) 123-4567


1. Enter a new Contact from the console
2. Import Contacts from a file
3. Display all Contacts on the console
4. Export all Contacts to a file
5. Delete a single Contact
6. Edit a single Contact
7. Display a selected subset of Contacts on the console
8. Export a selected subset of Contacts to a file
9. Exit
Enter choice: 9

HW4 Complete
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Conclusion (1 paragraph)

The project was successful because all the requirements were met and all the menu options work correctly. The code is organized in a way that is easy to understand. I learned that BSTs automatically maintain sorted order and separating concerns into different classes make code easier to test and understand. In the future, I would use this design approach.